

Core Java Programs 21-30

21. Check Leap Year

```
class LeapYear {
    public static void main(String[] args) {
        int year = 2024;
        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
            System.out.println(year + " is a Leap Year");
        else
            System.out.println(year + " is Not a Leap Year");
    }
}
```

22. Calculate Power of a Number

```
class Power {
    public static void main(String[] args) {
        int base = 2, exponent = 5, result = 1;
        for (int i = 1; i <= exponent; i++)
            result *= base;
        System.out.println("Power = " + result);
    }
}
```

23. Print Numbers from 1 to 100

```
class PrintNumbers {
    public static void main(String[] args) {
        for (int i = 1; i <= 100; i++)
            System.out.println(i);
    }
}
```

24. Sum of First N Natural Numbers

```
class SumN {
    public static void main(String[] args) {
        int n = 10, sum = 0;
        for (int i = 1; i <= n; i++)
            sum += i;
        System.out.println("Sum = " + sum);
    }
}
```

25. Average of Array Elements

```
class AverageArray {
    public static void main(String[] args) {
        int arr[] = {10, 20, 30, 40, 50};
        int sum = 0;
        for (int i = 0; i < arr.length; i++)
            sum += arr[i];
        double average = (double) sum / arr.length;
        System.out.println("Average = " + average);
    }
}
```

26. Copy One Array into Another Array

```
class CopyArray {
    public static void main(String[] args) {
        int source[] = {1, 2, 3, 4, 5};
        int destination[] = new int[source.length];
    }
}
```

```

        for(int i=0;i<source.length;i++)
            destination[i] = source[i];
        for(int i=0;i<destination.length;i++)
            System.out.print(destination[i] + " ");
    }
}

```

27. Count Vowels in a String

```

class CountVowels {
    public static void main(String[] args) {
        String str = "Java Programming";
        int count = 0;
        str = str.toLowerCase();
        for(int i=0;i<str.length();i++){
            char ch = str.charAt(i);
            if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
                count++;
        }
        System.out.println("Number of Vowels = " + count);
    }
}

```

28. Convert String to Uppercase

```

class UpperCase {
    public static void main(String[] args) {
        String str = "java programming";
        System.out.println(str.toUpperCase());
    }
}

```

29. Check String Palindrome

```

class StringPalindrome {
    public static void main(String[] args) {
        String str = "madam";
        String reverse = "";
        for(int i=str.length()-1;i>=0;i--){
            reverse += str.charAt(i);
        }
        if(str.equals(reverse))
            System.out.println("Palindrome String");
        else
            System.out.println("Not a Palindrome String");
    }
}

```

30. Sort Array in Ascending Order

```

class SortArray {
    public static void main(String[] args) {
        int arr[] = {5,2,8,1,9};
        for(int i=0;i<arr.length;i++){
            for(int j=i+1;j<arr.length;j++){
                if(arr[i] > arr[j]){
                    int temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        for(int i=0;i<arr.length;i++)
            System.out.print(arr[i] + " ");
    }
}

```